Quantium Project

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My Key Findings:

Young Singles/Couples generated the highest chip sales. Mainstream customers purchased the largest quantity. 175g pack sizes were most popular. Brand X dominated premium segments.

My Recommendation: The company should target Mainstream Young Singles/Couples. Increase promotions on popular pack sizes. Place premium brands near high-traffic areas.

```
Behaviour_data <- read.csv("QVI_purchase_behaviour.csv")

head(Behaviour_data)
```

```
##   LYLTY_CARD_NBR              LIFESTAGE PREMIUM_CUSTOMER
## 1           1000  YOUNG SINGLES/COUPLES          Premium
## 2           1002  YOUNG SINGLES/COUPLES       Mainstream
## 3           1003         YOUNG FAMILIES           Budget
## 4           1004  OLDER SINGLES/COUPLES       Mainstream
## 5           1005 MIDAGE SINGLES/COUPLES       Mainstream
## 6           1007  YOUNG SINGLES/COUPLES           Budget
```

package installation

install.packages("tidyverse", repos = "https://r-project.org") install.packages("janitor") install.packages("lubridate") install.packages("stringr")

Import excel file(install package first)

install.packages("readxl")

```
library(readxl)
Transaction_data <- read_excel("QVI_transaction_data.xlsx")

head(Transaction_data)
```

```
## # A tibble: 6 × 8
##    DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME    PROD_QTY TOT
_SALES
##   <dbl>     <dbl>          <dbl>  <dbl>    <dbl> <chr>           <dbl>
 <dbl>
## 1 43390         1           1000      1        5 Natural Chi…        2
   6
## 2 43599         1           1307    348       66 CCs Nacho C…        3
   6.3
## 3 43605         1           1343    383       61 Smiths Crin…        2
```

```
   2.9
## 4 43329         2           2373    974       69 Smiths Chip…        5
  15
## 5 43330         2           2426   1038      108 Kettle Tort…        3
  13.8
## 6 43604         4           4074   2982       57 Old El Paso…        1
   5.1
```

```r
str(Behaviour_data)
```

```
## 'data.frame':    72637 obs. of  3 variables:
##  $ LYLTY_CARD_NBR  : int  1000 1002 1003 1004 1005 1007 1009 1010 1011 101
2 ...
##  $ LIFESTAGE       : chr  "YOUNG SINGLES/COUPLES" "YOUNG SINGLES/COUPLES"
"YOUNG FAMILIES" "OLDER SINGLES/COUPLES" ...
##  $ PREMIUM_CUSTOMER: chr  "Premium" "Mainstream" "Budget" "Mainstream" ...
```

```r
str(Transaction_data)
```

```
## tibble [264,836 × 8] (S3: tbl_df/tbl/data.frame)
##  $ DATE          : num [1:264836] 43390 43599 43605 43329 43330 ...
##  $ STORE_NBR     : num [1:264836] 1 1 1 2 2 4 4 4 5 7 ...
##  $ LYLTY_CARD_NBR: num [1:264836] 1000 1307 1343 2373 2426 ...
##  $ TXN_ID        : num [1:264836] 1 348 383 974 1038 ...
##  $ PROD_NBR      : num [1:264836] 5 66 61 69 108 57 16 24 42 52 ...
##  $ PROD_NAME     : chr [1:264836] "Natural Chip        Compny SeaSalt175g"
 "CCs Nacho Cheese    175g" "Smiths Crinkle Cut  Chips Chicken 170g" "Smiths
Chip Thinly  S/Cream&Onion 175g" ...
##  $ PROD_QTY      : num [1:264836] 2 3 2 5 3 1 1 1 1 2 ...
##  $ TOT_SALES     : num [1:264836] 6 6.3 2.9 15 13.8 5.1 5.7 3.6 3.9 7.2 ...
```

Summary statistics

```r
summary(Behaviour_data)
```

```
##  LYLTY_CARD_NBR       LIFESTAGE       PREMIUM_CUSTOMER
##  Min.   :   1000   Length    :72637   Length    :72637
##  1st Qu.:  66202   N.unique :     7   N.unique :     3
##  Median : 134040   N.blank  :     0   N.blank  :     0
##  Mean   : 136186   Min.nchar:     8   Min.nchar:     6
##  3rd Qu.: 203375   Max.nchar:    22   Max.nchar:    10
##  Max.   :2373711
```

```r
summary(Transaction_data)
```

```
##       DATE         STORE_NBR     LYLTY_CARD_NBR        TXN_ID
##  Min.   :43282   Min.   :  1.0   Min.   :   1000   Min.   :      1
##  1st Qu.:43373   1st Qu.: 70.0   1st Qu.:  70021   1st Qu.:  67602
##  Median :43464   Median :130.0   Median : 130358   Median : 135138
##  Mean   :43464   Mean   :135.1   Mean   : 135549   Mean   : 135158
##  3rd Qu.:43555   3rd Qu.:203.0   3rd Qu.: 203094   3rd Qu.: 202701
##  Max.   :43646   Max.   :272.0   Max.   :2373711   Max.   :2415841
```

```
##     PROD_NBR          PROD_NAME           PROD_QTY          TOT_SALES
##  Min.   :  1.00   Length   :264836   Min.   :  1.000   Min.   :  1.500
##  1st Qu.: 28.00   N.unique :   114   1st Qu.:  2.000   1st Qu.:  5.400
##  Median : 56.00   N.blank  :     0   Median :  2.000   Median :  7.400
##  Mean   : 56.58   Min.nchar:    17   Mean   :  1.907   Mean   :  7.304
##  3rd Qu.: 85.00   Max.nchar:    40   3rd Qu.:  2.000   3rd Qu.:  9.200
##  Max.   :114.00                      Max.   :200.000   Max.   :650.000
```

Check for missing values

```
colSums(is.na(Transaction_data))

##           DATE      STORE_NBR LYLTY_CARD_NBR         TXN_ID       PROD_NBR

##              0              0              0              0              0

##      PROD_NAME       PROD_QTY      TOT_SALES
##              0              0              0

colSums(is.na(Behaviour_data))

##   LYLTY_CARD_NBR        LIFESTAGE PREMIUM_CUSTOMER
##                0                0                0
```

No missing values in both data

Checking for duplicates

```
sum(duplicated(Transaction_data))

## [1] 1

sum(duplicated(Behaviour_data))

## [1] 0
```

Removing duplicates

```
library(tidyverse)

## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
```

```
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
 conflicts to become errors
```

```r
Transaction_data <- distinct(Transaction_data)
sum(duplicated(Transaction_data))
```

```
## [1] 0
```

Converting Excel numeric date to actual date

```r
Transaction_data$DATE <- as.Date(Transaction_data$DATE, origin = "1899-12-30")
head(Transaction_data$DATE)
```

```
## [1] "2018-10-17" "2019-05-14" "2019-05-20" "2018-08-17" "2018-08-18"
## [6] "2019-05-19"
```

```r
unique(Transaction_data$PROD_NAME)
```

```
##   [1] "Natural Chip        Compny SeaSalt175g"
##   [2] "CCs Nacho Cheese    175g"
##   [3] "Smiths Crinkle Cut  Chips Chicken 170g"
##   [4] "Smiths Chip Thinly  S/Cream&Onion 175g"
##   [5] "Kettle Tortilla ChpsHny&Jlpno Chili 150g"
##   [6] "Old El Paso Salsa   Dip Tomato Mild 300g"
##   [7] "Smiths Crinkle Chips Salt & Vinegar 330g"
##   [8] "Grain Waves         Sweet Chilli 210g"
##   [9] "Doritos Corn Chip Mexican Jalapeno 150g"
##  [10] "Grain Waves Sour    Cream&Chives 210G"
##  [11] "Kettle Sensations   Siracha Lime 150g"
##  [12] "Twisties Cheese     270g"
##  [13] "WW Crinkle Cut      Chicken 175g"
##  [14] "Thins Chips Light&  Tangy 175g"
##  [15] "CCs Original 175g"
##  [16] "Burger Rings 220g"
##  [17] "NCC Sour Cream &    Garden Chives 175g"
##  [18] "Doritos Corn Chip Southern Chicken 150g"
##  [19] "Cheezels Cheese Box 125g"
##  [20] "Smiths Crinkle      Original 330g"
##  [21] "Infzns Crn Crnchers Tangy Gcamole 110g"
##  [22] "Kettle Sea Salt     And Vinegar 175g"
##  [23] "Smiths Chip Thinly  Cut Original 175g"
##  [24] "Kettle Original 175g"
##  [25] "Red Rock Deli Thai  Chilli&Lime 150g"
##  [26] "Pringles Sthrn FriedChicken 134g"
##  [27] "Pringles Sweet&Spcy BBQ 134g"
##  [28] "Red Rock Deli SR    Salsa & Mzzrlla 150g"
##  [29] "Thins Chips         Originl saltd 175g"
##  [30] "Red Rock Deli Sp    Salt & Truffle 150G"
##  [31] "Smiths Thinly       Swt Chli&S/Cream175G"
##  [32] "Kettle Chilli 175g"
##  [33] "Doritos Mexicana    170g"
```

```
##  [34] "Smiths Crinkle Cut  French OnionDip 150g"
##  [35] "Natural ChipCo      Hony Soy Chckn175g"
##  [36] "Dorito Corn Chp     Supreme 380g"
##  [37] "Twisties Chicken270g"
##  [38] "Smiths Thinly Cut   Roast Chicken 175g"
##  [39] "Smiths Crinkle Cut  Tomato Salsa 150g"
##  [40] "Kettle Mozzarella   Basil & Pesto 175g"
##  [41] "Infuzions Thai SweetChili PotatoMix 110g"
##  [42] "Kettle Sensations   Camembert & Fig 150g"
##  [43] "Smith Crinkle Cut   Mac N Cheese 150g"
##  [44] "Kettle Honey Soy    Chicken 175g"
##  [45] "Thins Chips Seasonedchicken 175g"
##  [46] "Smiths Crinkle Cut  Salt & Vinegar 170g"
##  [47] "Infuzions BBQ Rib   Prawn Crackers 110g"
##  [48] "GrnWves Plus Btroot & Chilli Jam 180g"
##  [49] "Tyrrells Crisps     Lightly Salted 165g"
##  [50] "Kettle Sweet Chilli And Sour Cream 175g"
##  [51] "Doritos Salsa       Medium 300g"
##  [52] "Kettle 135g Swt Pot Sea Salt"
##  [53] "Pringles SourCream  Onion 134g"
##  [54] "Doritos Corn Chips  Original 170g"
##  [55] "Twisties Cheese     Burger 250g"
##  [56] "Old El Paso Salsa   Dip Chnky Tom Ht300g"
##  [57] "Cobs Popd Swt/Chlli &Sr/Cream Chips 110g"
##  [58] "Woolworths Mild     Salsa 300g"
##  [59] "Natural Chip Co     Tmato Hrb&Spce 175g"
##  [60] "Smiths Crinkle Cut  Chips Original 170g"
##  [61] "Cobs Popd Sea Salt  Chips 110g"
##  [62] "Smiths Crinkle Cut  Chips Chs&Onion170g"
##  [63] "French Fries Potato Chips 175g"
##  [64] "Old El Paso Salsa   Dip Tomato Med 300g"
##  [65] "Doritos Corn Chips  Cheese Supreme 170g"
##  [66] "Pringles Original   Crisps 134g"
##  [67] "RRD Chilli&         Coconut 150g"
##  [68] "WW Original Corn    Chips 200g"
##  [69] "Thins Potato Chips  Hot & Spicy 175g"
##  [70] "Cobs Popd Sour Crm  &Chives Chips 110g"
##  [71] "Smiths Crnkle Chip  Orgnl Big Bag 380g"
##  [72] "Doritos Corn Chips  Nacho Cheese 170g"
##  [73] "Kettle Sensations   BBQ&Maple 150g"
##  [74] "WW D/Style Chip     Sea Salt 200g"
##  [75] "Pringles Chicken    Salt Crips 134g"
##  [76] "WW Original Stacked Chips 160g"
##  [77] "Smiths Chip Thinly  CutSalt/Vinegr175g"
##  [78] "Cheezels Cheese 330g"
##  [79] "Tostitos Lightly    Salted 175g"
##  [80] "Thins Chips Salt &  Vinegar 175g"
##  [81] "Smiths Crinkle Cut  Chips Barbecue 170g"
##  [82] "Cheetos Puffs 165g"
##  [83] "RRD Sweet Chilli &  Sour Cream 165g"
```

```
##  [84] "WW Crinkle Cut      Original 175g"
##  [85] "Tostitos Splash Of  Lime 175g"
##  [86] "Woolworths Medium   Salsa 300g"
##  [87] "Kettle Tortilla ChpsBtroot&Ricotta 150g"
##  [88] "CCs Tasty Cheese    175g"
##  [89] "Woolworths Cheese   Rings 190g"
##  [90] "Tostitos Smoked     Chipotle 175g"
##  [91] "Pringles Barbeque   134g"
##  [92] "WW Supreme Cheese   Corn Chips 200g"
##  [93] "Pringles Mystery    Flavour 134g"
##  [94] "Tyrrells Crisps     Ched & Chives 165g"
##  [95] "Snbts Whlgrn Crisps Cheddr&Mstrd 90g"
##  [96] "Cheetos Chs & Bacon Balls 190g"
##  [97] "Pringles Slt Vingar 134g"
##  [98] "Infuzions SourCream&Herbs Veg Strws 110g"
##  [99] "Kettle Tortilla ChpsFeta&Garlic 150g"
## [100] "Infuzions Mango     Chutny Papadums 70g"
## [101] "RRD Steak &         Chimuchurri 150g"
## [102] "RRD Honey Soy       Chicken 165g"
## [103] "Sunbites Whlegrn    Crisps Frch/Onin 90g"
## [104] "RRD Salt & Vinegar  165g"
## [105] "Doritos Cheese      Supreme 330g"
## [106] "Smiths Crinkle Cut  Snag&Sauce 150g"
## [107] "WW Sour Cream &OnionStacked Chips 160g"
## [108] "RRD Lime & Pepper   165g"
## [109] "Natural ChipCo Sea  Salt & Vinegr 175g"
## [110] "Red Rock Deli Chikn&Garlic Aioli 150g"
## [111] "RRD SR Slow Rst     Pork Belly 150g"
## [112] "RRD Pc Sea Salt     165g"
## [113] "Smith Crinkle Cut   Bolognese 150g"
## [114] "Doritos Salsa Mild  300g"
```

Count products containing salsa and Filter out salsa products

```
Transaction_data %>%
  filter(str_detect(PROD_NAME, "salsa|Salsa"))
```

```
## # A tibble: 18,094 × 8
##    DATE       STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME        PR
OD_QTY
##    <date>         <dbl>          <dbl>  <dbl>    <dbl> <chr>
 <dbl>
##  1 2019-05-19         4           4074   2982       57 Old El Paso Sal…
     1
##  2 2019-05-15        39          39144  35506       57 Old El Paso Sal…
     1
##  3 2019-05-20        45          45127  41122       64 Red Rock Deli S…
     2
##  4 2018-08-18        56          56013  50090       39 Smiths Crinkle …
     1
```

```
## 5 2019-05-15 82 82480 82047 101 Doritos Salsa …
1
## 6 2018-08-15 94 94233 93956 65 Old El Paso Sal…
1
## 7 2018-08-16 97 97159 97271 35 Woolworths Mild…
5
## 8 2018-08-15 116 116184 120270 59 Old El Paso Sal…
1
## 9 2018-08-16 157 157185 159562 59 Old El Paso Sal…
2
## 10 2018-08-19 175 175306 176634 59 Old El Paso Sal…
1
## # ℹ 18,084 more rows
## # ℹ 1 more variable: TOT_SALES <dbl>
```

```
Transaction_data <- Transaction_data %>%
  filter(!str_detect(PROD_NAME, "salsa|Salsa"))

Transaction_data <- Transaction_data %>%
  mutate(PACK_SIZE = str_extract(PROD_NAME, "\\d+")) %>%
  mutate(PACK_SIZE = as.numeric(PACK_SIZE))


head(Transaction_data$PACK_SIZE)
```

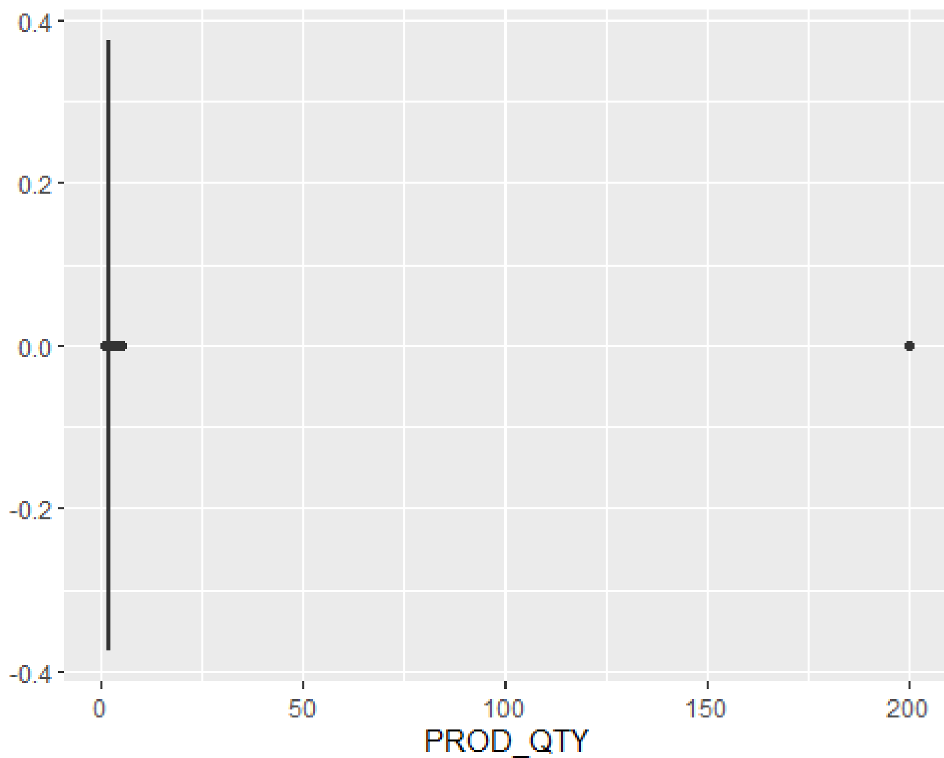
```
## [1] 175 175 170 175 150 330
```

```
Transaction_data <- Transaction_data %>%
  mutate(BRAND = word(PROD_NAME, 1))

unique(Transaction_data$BRAND)
```

```
##  [1] "Natural"    "CCs"        "Smiths"     "Kettle"     "Grain"
##  [6] "Doritos"    "Twisties"   "WW"         "Thins"      "Burger"
## [11] "NCC"        "Cheezels"   "Infzns"     "Red"        "Pringles"
## [16] "Dorito"     "Infuzions"  "Smith"      "GrnWves"    "Tyrrells"
## [21] "Cobs"       "French"     "RRD"        "Tostitos"   "Cheetos"
## [26] "Woolworths" "Snbts"      "Sunbites"
```

Fixing inconsistent spellings

```
Transaction_data$BRAND <- recode(Transaction_data$BRAND,
                             "RED" = "RRD")


Transaction_data %>%
  ggplot(aes(x = PROD_QTY)) +
  geom_boxplot()
```




```
head(Transaction_data)

## # A tibble: 6 × 10
##   DATE       STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME          PR
OD_QTY
##   <date>         <dbl>          <dbl>  <dbl>    <dbl> <chr>
 <dbl>
## 1 2018-10-17         1           1000      1        5 Natural Chip     …
    2
## 2 2019-05-14         1           1307    348       66 CCs Nacho Cheese…
    3
## 3 2019-05-20         1           1343    383       61 Smiths Crinkle C…
    2
## 4 2018-08-17         2           2373    974       69 Smiths Chip Thin…
    5
## 5 2018-08-18         2           2426   1038      108 Kettle Tortilla …
    3
## 6 2019-05-16         4           4149   3333       16 Smiths Crinkle C…
    1
## # ℹ 3 more variables: TOT_SALES <dbl>, PACK_SIZE <dbl>, BRAND <chr>
```

Customers purchasing unusually high quantities and remove outlier Customers

```
Transaction_data %>%
  group_by(LYLTY_CARD_NBR) %>%
  summarise(total_qty = sum(PROD_QTY)) %>%
  arrange(desc(total_qty))
```

```
## # A tibble: 71,288 × 2
##    LYLTY_CARD_NBR total_qty
##             <dbl>     <dbl>
##  1         226000       400
##  2         230078        36
##  3         162039        34
##  4         113080        32
##  5         116181        32
##  6         179228        32
##  7          23192        31
##  8         105026        31
##  9         128178        31
## 10         138222        31
## # ℹ 71,278 more rows
```

```
Transaction_data <- Transaction_data %>%
  filter(LYLTY_CARD_NBR != 226000)
```

Merge data

```
library("tidyverse")
merged_data <- Transaction_data %>%
  left_join(Behaviour_data, by = "LYLTY_CARD_NBR")

head(merged_data)
```

```
## # A tibble: 6 × 12
##   DATE       STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME          PR
OD_QTY
##   <date>         <dbl>          <dbl>  <dbl>    <dbl> <chr>
 <dbl>
## 1 2018-10-17         1           1000      1        5 Natural Chip    …
     2
## 2 2019-05-14         1           1307    348       66 CCs Nacho Cheese…
     3
## 3 2019-05-20         1           1343    383       61 Smiths Crinkle C…
     2
## 4 2018-08-17         2           2373    974       69 Smiths Chip Thin…
     5
## 5 2018-08-18         2           2426   1038      108 Kettle Tortilla …
     3
## 6 2019-05-16         4           4149   3333       16 Smiths Crinkle C…
     1
## # ℹ 5 more variables: TOT_SALES <dbl>, PACK_SIZE <dbl>, BRAND <chr>,
## #   LIFESTAGE <chr>, PREMIUM_CUSTOMER <chr>
```

merged_data <- merged_data %>% mutate(TOT_SALES = PROD_QTY * TOT_SALES)

Analyzing Sales by Customer Segment

```r
segment_sales <- merged_data %>%
  group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
  summarise(
    total_sales = sum(TOT_SALES),
    total_customers = n_distinct(LYLTY_CARD_NBR),
    avg_sales = mean(TOT_SALES)
  )
```

```
## `summarise()` has regrouped the output.
## ℹ Summaries were computed grouped by LIFESTAGE and PREMIUM_CUSTOMER.
## ℹ Output is grouped by LIFESTAGE.
## ℹ Use `summarise(.groups = "drop_last")` to silence this message.
## ℹ Use `summarise(.by = c(LIFESTAGE, PREMIUM_CUSTOMER))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```r
segment_sales
```

```
## # A tibble: 21 × 5
## # Groups:   LIFESTAGE [7]
##    LIFESTAGE              PREMIUM_CUSTOMER total_sales total_customers avg
_sales
##    <chr>                  <chr>                  <dbl>           <int>
 <dbl>
##  1 MIDAGE SINGLES/COUPLES Budget                33346.            1474
  7.11
##  2 MIDAGE SINGLES/COUPLES Mainstream            84734.            3298
  7.64
##  3 MIDAGE SINGLES/COUPLES Premium               54444.            2369
  7.15
##  4 NEW FAMILIES           Budget                20607.            1087
  7.30
##  5 NEW FAMILIES           Mainstream            15980.             830
  7.31
##  6 NEW FAMILIES           Premium               10761.             575
  7.23
##  7 OLDER FAMILIES         Budget               156864.            4611
  7.29
##  8 OLDER FAMILIES         Mainstream            96414.            2788
  7.28
##  9 OLDER FAMILIES         Premium               75243.            2231
  7.23
## 10 OLDER SINGLES/COUPLES  Budget               127834.            4849
  7.44
## # ℹ 11 more rows
```

Top-Selling Brands

```r
brand_sales <- merged_data %>%
  group_by(BRAND) %>%
  summarise(total_sales = sum(TOT_SALES)) %>%
  arrange(desc(total_sales))
```

```
brand_sales
```

```
## # A tibble: 28 × 2
##    BRAND     total_sales
##    <chr>           <dbl>
##  1 Kettle        390240.
##  2 Smiths        202903.
##  3 Doritos       187278.
##  4 Pringles      177656.
##  5 Thins          88853.
##  6 Twisties       81522.
##  7 Tostitos       79790.
##  8 Infuzions      76248.
##  9 Cobs           70570.
## 10 RRD            64954.
## # ℹ 18 more rows
```

Analyzing Pack Size Preferences

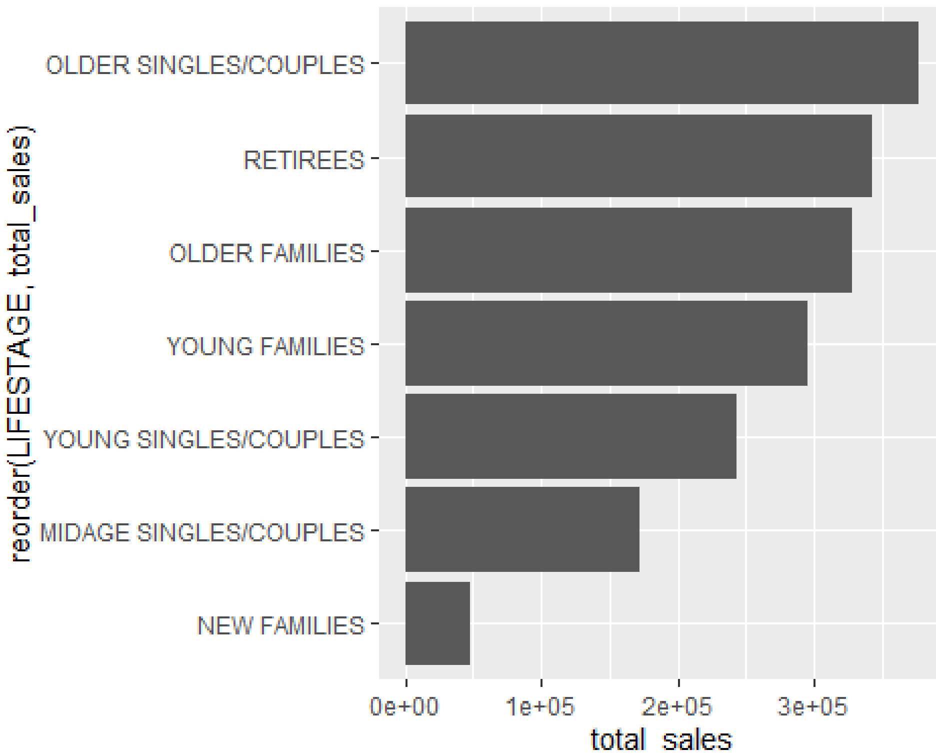
```
pack_sales <- merged_data %>%
  group_by(PACK_SIZE) %>%
  summarise(total_sales = sum(TOT_SALES)) %>%
  arrange(desc(total_sales))

pack_sales
```

```
## # A tibble: 20 × 2
##    PACK_SIZE total_sales
##        <dbl>       <dbl>
##  1       175     485431.
##  2       150     289682.
##  3       134     177656.
##  4       110     162765.
##  5       170     146673
##  6       330     136794.
##  7       165     101361.
##  8       380      75420.
##  9       270      55425.
## 10       210      43049.
## 11       250      26097.
## 12       135      26090.
## 13       200      16008.
## 14       190      14413.
## 15       160      10648.
## 16        90       9676.
## 17       180       8568.
## 18        70       6852
## 19       220       6831
## 20       125       5733
```

Visuals

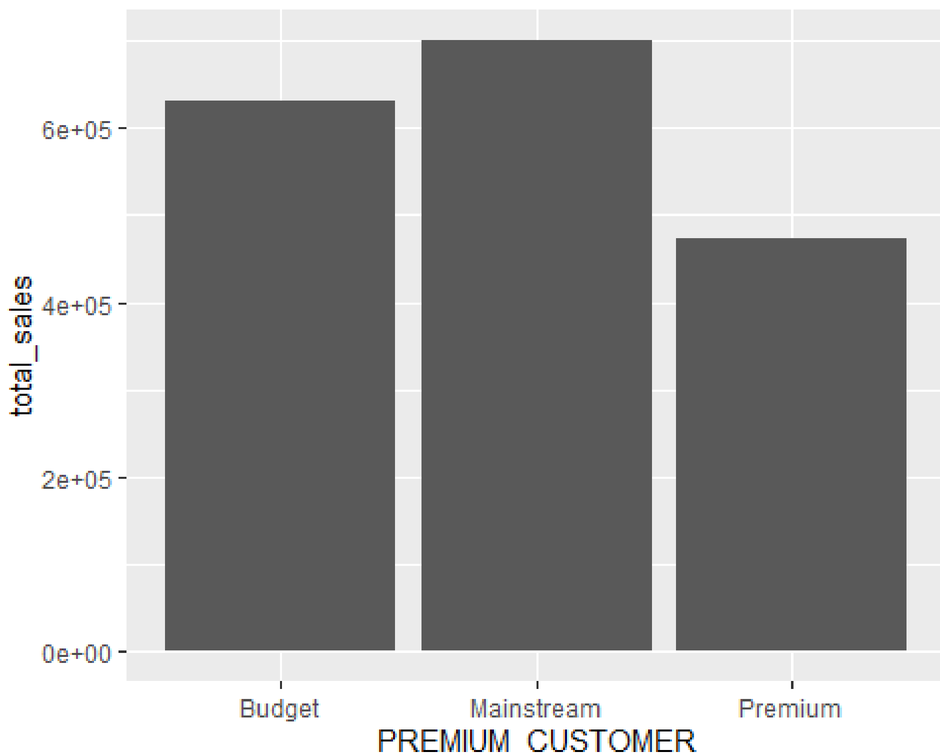
Sales by Life Stage

```
merged_data %>%
  group_by(LIFESTAGE) %>%
  summarise(total_sales = sum(TOT_SALES)) %>%
  ggplot(aes(x = reorder(LIFESTAGE, total_sales),
             y = total_sales)) +
  geom_col() +
  coord_flip()
```



Sales by Premium Segment

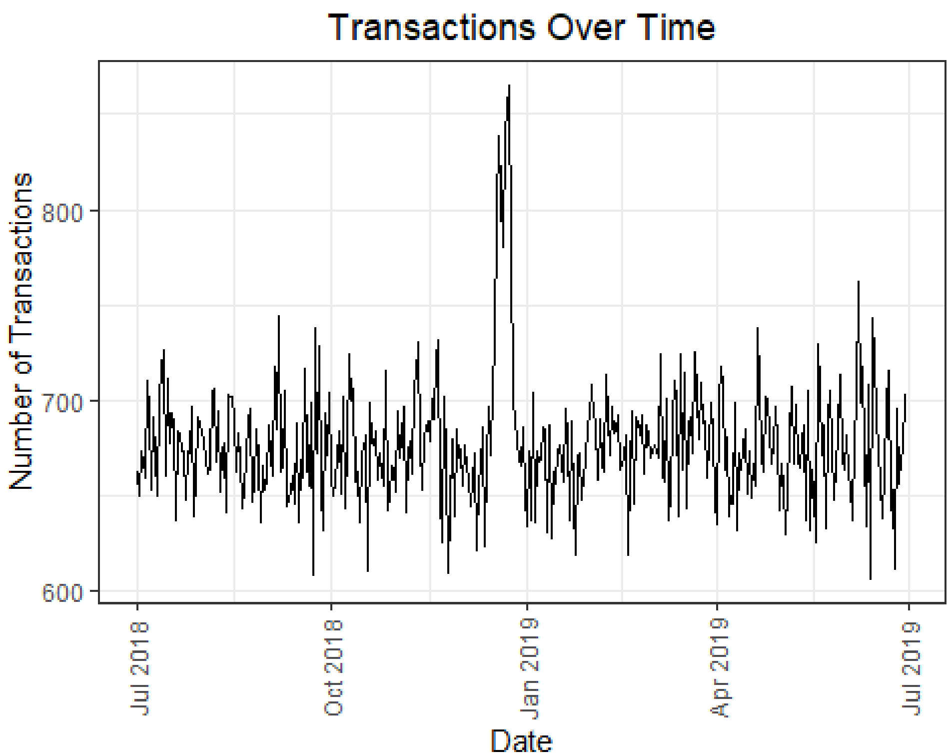
```
merged_data %>%
  group_by(PREMIUM_CUSTOMER) %>%
  summarise(total_sales = sum(TOT_SALES)) %>%
  ggplot(aes(x = PREMIUM_CUSTOMER,
             y = total_sales)) +
  geom_col()
```




```
theme_set(theme_bw())
theme_update(plot.title = element_text(hjust = 0.5))
```

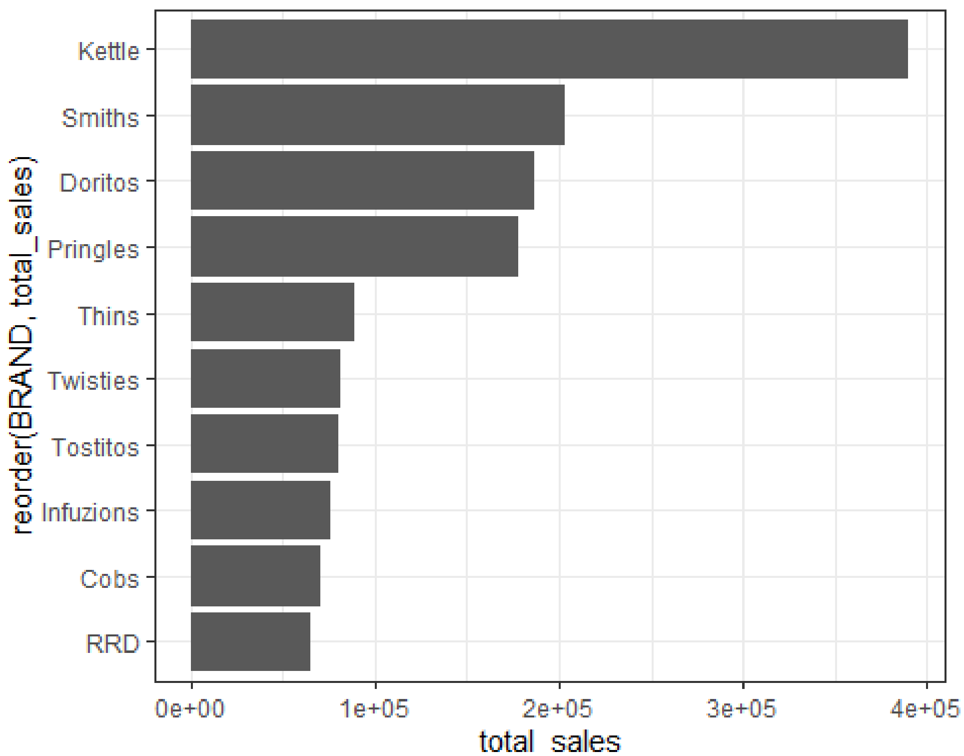
Plot transactions over time

```
transactions_over_time <- Transaction_data %>%
  group_by(DATE) %>%
  summarise(num_transactions = n())

transactions_over_time %>%
  ggplot(aes(x = DATE, y = num_transactions)) +
  geom_line() +
  labs(
    title = "Transactions Over Time",
    x = "Date",
    y = "Number of Transactions"
  ) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5))
```



Top Brands

```
brand_sales %>%
  top_n(10, total_sales) %>%
  ggplot(aes(x = reorder(BRAND, total_sales),
             y = total_sales)) +
  geom_col() +
  coord_flip()
```


Kettle
Smiths
Doritos
Pringles
Thins
Twisties
Tostitos
Infuzions
Cobs
RRD
reorder(BRAND, total_sales)
0e+00
1e+05
2e+05
3e+05
4e+05
total_sales